E4.1.1 Accommodation of the eye

Accommodation changes the crystalline lens's shape and focal length to focus objects at different distances on the approximately fixed retina. The cornea provides most fixed refraction, while the lens provides adjustment.

Near objects need stronger convergence. The ciliary muscle contracts, reducing suspensory-ligament tension. The lens thickens, focal length decreases and optical power increases. For distant objects, the muscle relaxes, ligament tension rises and the lens is pulled thinner. Focal length increases and optical power decreases.

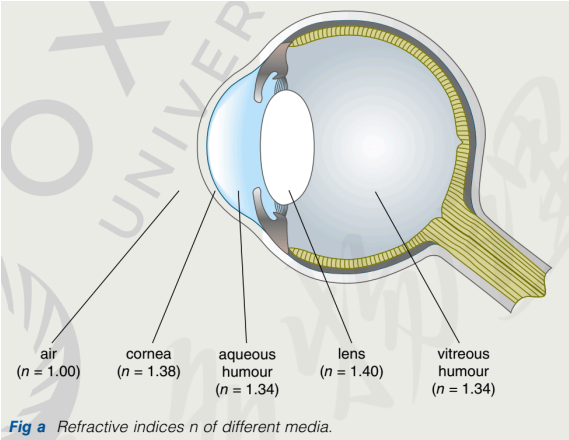
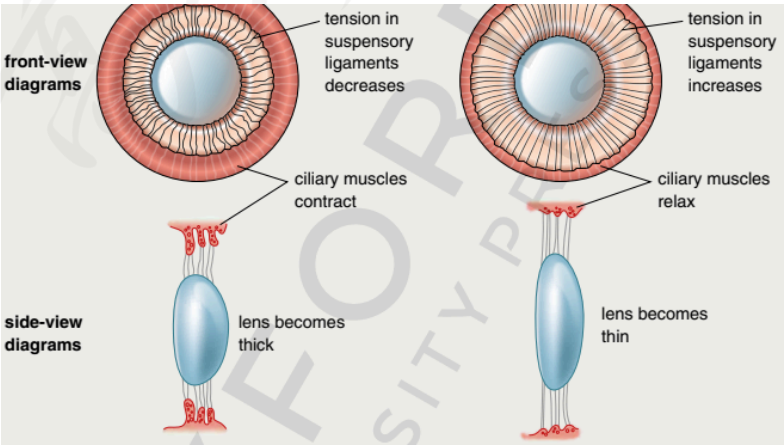


Fig a *Refractive indices n of different media.*

The cornea and lens refract light; accommodation changes the lens shape while the retina distance remains almost fixed.



Left: near vision, with contracted ciliary muscles and a thicker lens. Right: distant vision, with relaxed muscles and a thinner lens.

E4.1.2 Corneal refraction and accommodation

The air-cornea boundary provides most of the eye's refraction and is not adjustable. The crystalline lens changes shape and focal length to focus objects at different distances on the retina.

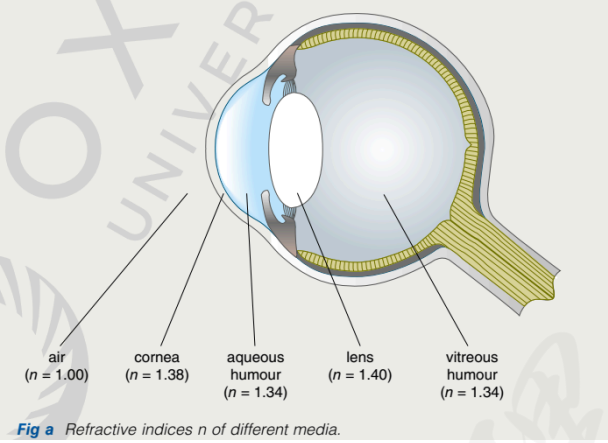
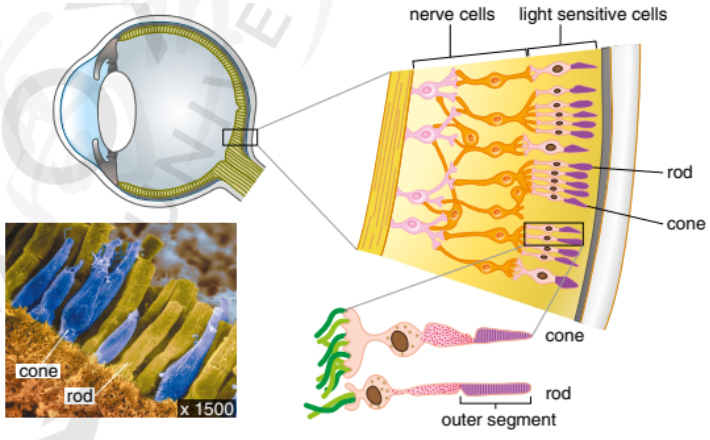


Fig a Refractive indices n of different media.

The cornea provides most refraction; the lens adjusts focal length for accommodation.

E4.1.3 Rods and cones

Rods are more sensitive in dim light and mainly support low-light, non-colour vision. Cones provide colour and fine detail under brighter conditions. Colour discrimination is therefore poorer in dim light.



Retinal rods and cones differ in shape and visual function.

E4.1.4 Visual angle

Visual angle is the angle subtended by an object's ends at the eye, depending on size and distance. For small angles, size divided by distance gives radians. Moving the same object farther away reduces visual angle. Compare angular separation with the eye's or instrument's resolution limit to decide whether two points are distinguishable.

$$\theta \approx \frac{x}{u}$$

Small visual angle θ in radians; x: object size; u: viewing distance, both in the same length unit.

E4.1.5 Rayleigh's criterion and resolving power

A finite aperture turns each point source's image into a diffraction pattern with a central bright spot and surrounding bright and dark rings. Nearby point sources produce overlapping patterns.

Rayleigh's criterion defines just-resolved images when one central maximum falls on the other's first dark ring. For a circular aperture, the minimum resolvable angle θ_R increases with wavelength λ and decreases with diameter d. Shorter wavelength or a larger aperture resolves closer points. Magnifying an image and separating overlapping point images are different functions.

$$\theta_R \approx 1.22\frac{\lambda}{d}$$

Rayleigh criterion for a circular aperture. θ_R: minimum resolvable angular separation (rad); λ: wavelength (m); d: aperture diameter (m), or pupil diameter for the eye.

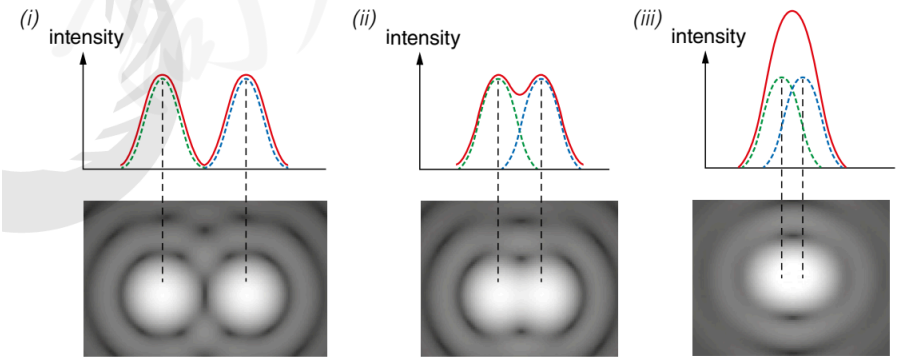


Fig 3.2d *According to Rayleigh criterion, two objects are (i) resolved clearly, (ii) just resolved, and (iii) not resolved.*

Compare clearly resolved, just resolved and unresolved pairs. At the Rayleigh limit, one central maximum falls on the first minimum of the other pattern.

E4.1.6 Optical power and lens combinations

Optical power is the reciprocal of focal length in metres, measured in dioptres. It is positive for converging and negative for diverging lenses. Shorter focal length means greater magnitude. Powers of thin lenses in contact add, but focal lengths do not. Appreciable separation requires a more complete combination calculation.

$$P = \frac{1}{f}$$

$$P_{\text{eff}} \approx P_1 + P_2 + \cdots$$

Optical power P in dioptres (D = m^{-1}); focal length f MUST be in metres. Converging lens: positive; diverging lens: negative.

Thin lenses in contact or with negligible separation; powers in dioptres. Significant separation requires an additional correction.

E4.1.7 Near point and far point

The near point is the closest position seen clearly with full accommodation. The far point is the farthest seen clearly with the eye relaxed. A normal far point is infinity, and about 25 cm is a common textbook near-point reference. Near point varies with age and individual, so is not a universal constant.

E4.1.8 Correcting myopia, hyperopia and presbyopia

A myopic eye focuses distant light too early and needs a concave lens to diverge it first. Hyperopia involves insufficient power or a shorter eyeball and can need a convex lens for near vision. Presbyopia is age-related loss of accommodation, often also corrected for near tasks with a convex lens, but has a different cause. Use the unaided near or far point to locate the needed virtual image.

How spectacles form a useful virtual image

Spectacles form a virtual image at a position the unaided eye can see clearly, such as its near or far point. The eye then focuses it onto the retina. Use positive real-object and negative same-side virtual-image distances in metres. Simple reciprocal-distance differences assume negligible lens-eye separation.

$$P_{\text{spectacles}} = \frac{1}{u} + \frac{1}{v}$$

$$P_{\text{spectacles}} = \frac{1}{d_{\text{corrected}}} - \frac{1}{d_{\text{uncorrected}}}$$

All distances in metres; power in D. Real object u = $d_{\text{corrected}} > 0$; virtual image at the unaided near/far point has v = $-d_{\text{uncorrected}}$. Use $1/\infty = 0$. Lens–eye separation neglected.

Combined power of eye and spectacles

In the thin-lens model of the eye, object distance is the target-to-eye distance and real-image distance is the optical-centre-to-retina distance. Clear focus requires these distances and optical power to satisfy the thin-lens relation. When spectacles are close to the eye, their powers approximately add. The retina's position remains fixed.

$$P_{\text{eye}} = \frac{1}{u} + \frac{1}{v}$$

$$P_{\text{eff}} = \frac{1}{u} + \frac{1}{v}$$

Unaided eye model: u: real-object distance at near/far point (m), positive; v: optical-centre-to-retina distance (m), positive; P: D.

Corrected eye plus spectacles treated as a combined thin lens: u: corrected near/far-point distance (m); v: retina distance (m); P_{eff}: D. Real object/image distances positive.

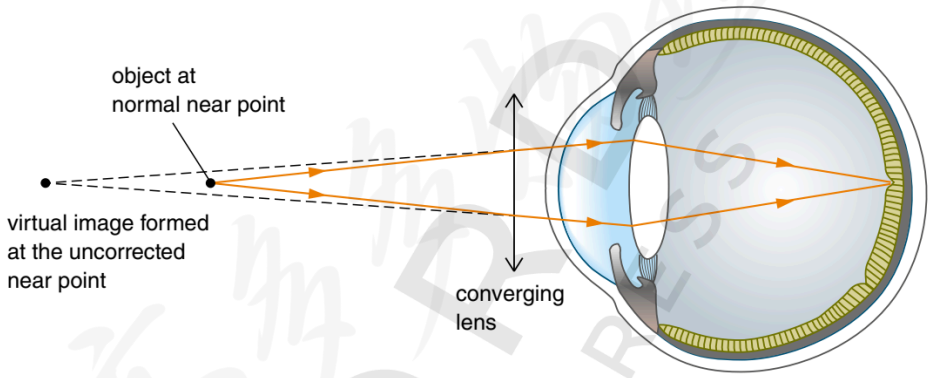


Fig 1.2h *Correcting old-sightedness with a converging lens.*

The converging spectacle lens produces a virtual image at the unaided near point. The eye then forms a real image on the retina; these are two different images.

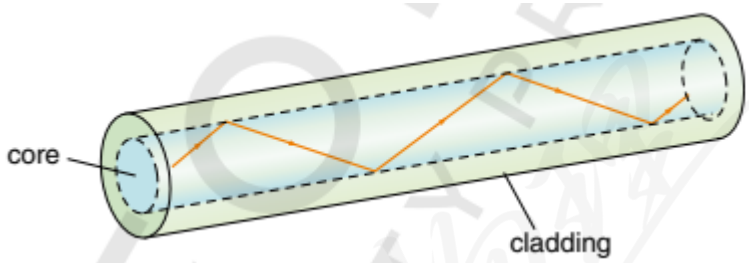
E4.1.9 Total internal reflection in fibres

A fibre's core has a higher index than its cladding. Light striking the internal boundary above the critical angle, measured from the normal, is totally internally reflected and guided along the core. Critical angle depends on the cladding-to-core index ratio. Not every incoming direction is trapped.

$$n_1 \sin C = n_2 \sin 90^\circ$$

$$\sin C = \frac{n_2}{n_1}$$

Core refractive index n_1 > cladding index n_2; C: critical angle from the normal. Total internal reflection requires incidence angle greater than C.



The core has a higher refractive index than the cladding, enabling total internal reflection at their boundary.

E4.1.10 Coherent and incoherent fibre bundles

A coherent image-transmitting bundle preserves the relative positions of fibres at its input and output. Each image point is therefore passed on in the original arrangement. A bundle used only for illumination need not preserve this arrangement. Here coherence refers to spatial ordering. In interference, it refers to a fixed phase relationship.

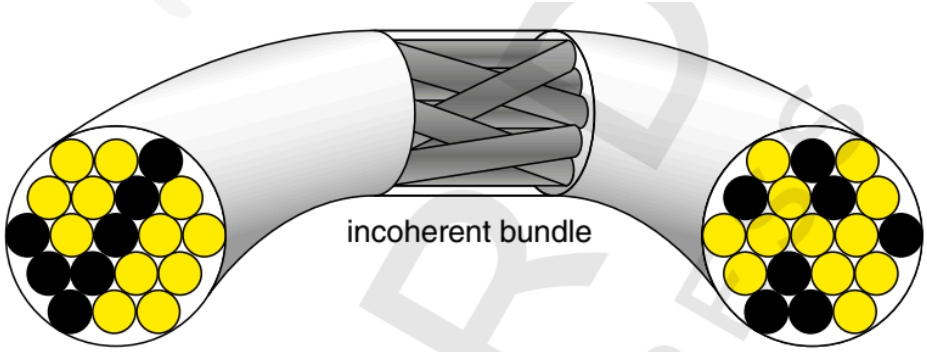


Fig 1.3j *An incoherent bundle fibre cannot be used to transmit images.*

An incoherent fibre bundle scrambles relative positions and carries illumination. Image transmission requires a coherent bundle with positions preserved.

E4.1.11 Advantages and limitations of fibre-optic endoscopy

Endoscopy directly views accessible internal organ surfaces and can pass small sampling tools through a channel. Its view and reach are limited. Some procedures require preparation such as fasting or anaesthesia, and risks include perforation or contamination.